

will have zero effect on the estimation error because the estimation error will be infinite regardless of the noise value. However, the estimate will obviously be poor. This example shows the importance of balancing H_∞ performance with other performance criteria.

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11.4 THE CONTINUOUS-TIME H_∞ FILTER

The methods of the earlier sections can also be used to derive a continuous-time H_∞ filter, as shown in [Rhe89, Ban91, Ban92]. In this section we consider the continuous-time system

$$\begin{aligned}\dot{x} &= Ax + Bu + w \\ y &= Cx + v \\ z &= Lx\end{aligned}\tag{11.115}$$

where L is a user-defined matrix and z is the vector that we want to estimate. Our estimate of z is denoted $\hat{z}$, and our estimate of the state at time 0 is denoted $\hat{x}(0)$. The vectors w and v are disturbances with unknown statistics; they may not even be zero-mean. In the game theory approach to H_∞ filtering we define the following cost function:

$$J_1 = \frac{\int_0^T \|z - \hat{z}\|_S^2 dt}{\|x(0) - \hat{x}(0)\|_{P_0}^2 + \int_0^T (\|w\|_{Q^{-1}}^2 + \|v\|_{R^{-1}}^2) dt}\tag{11.116}$$

P_0 , Q , R , and S are positive definite matrices chosen by the engineer based on the specific problem. Our goal is to find an estimator such that

$$J_1 < \frac{1}{\theta}\tag{11.117}$$

The estimator that solves this problem is given by

$$\begin{aligned}P(0) &= P_0 \\ \dot{P} &= AP + PA^T + Q - KCP + \theta PL^T SLP \\ K &= PC^T R^{-1} \\ \dot{\hat{x}} &= A\hat{x} + Bu + K(y - C\hat{x}) \\ \hat{z} &= L\hat{x}\end{aligned}\tag{11.118}$$

These equations are identical to the continuous-time Kalman filter equations (see Section 8.2) except for the θ term in the $\dot{P}$ equation. The inclusion of the θ term in the $\dot{P}$ equation tends to increase P , which tends to increase the gain K , which tends to make the estimator more responsive to measurements than the Kalman filter. This is a way of robustifying the filter to uncertainty in the system model. The estimator given above solves the H_∞ estimation problem if and only if $P(t)$ remains positive definite for all $t \in [0, T]$. As with the discrete-time filter, we can also obtain a steady-state continuous-time H_∞ filter. To do this we let $\dot{P} = 0$ so that the differential Riccati equation above reduces to an algebraic Riccati equation.

■ **EXAMPLE 11.4**

Consider the scalar continuous-time system

$$\begin{aligned}\dot{x} &= x + w \\ y &= x + v \\ z &= x\end{aligned}\tag{11.119}$$

We see that $A = C = L = 1$. Further suppose that $Q = R = S = 1$ in the cost function of Equation (11.116). Then the differential Riccati equation for the H_∞ filter is

$$\begin{aligned}\dot{P} &= AP + PA^T + Q - PC^T R^{-1} CP + \theta PL^T SLP \\ &= 2P + 1 + (\theta - 1)P^2\end{aligned}\tag{11.120}$$

This can be solved numerically or analytically as a function of time for a given θ to give P , and then the H_∞ gain $K = PC^T R^{-1} = P$ can be obtained. We can also set $\dot{P} = 0$ in Equation (11.120) to obtain the steady-state solution for P . This gives

$$(\theta - 1)P^2 + 2P + 1 = 0\tag{11.121}$$

As mentioned above, the solution to this quadratic equation must be positive definite in order for it to solve the H_∞ estimation problem. For this scalar equation, positive definite simply means positive. The equation has a positive solution for $\theta < 1$, in which case the steady-state solution is given by

$$P = \frac{-1 - \sqrt{2 - \theta}}{\theta - 1}\tag{11.122}$$

Suppose we choose $\theta = 7/16$. In this case, the analytic solution for the time-varying P can be obtained from Equation (11.120) as

$$\begin{aligned}P(t) &= \frac{4 + 160ce^{5t/2}}{-9 + 40ce^{5t/2}} \\ c &= \frac{9P(0) + 4}{40P(0) - 160}\end{aligned}\tag{11.123}$$

From this analytic expression for $P(t)$ we can see that

$$\lim_{t \rightarrow \infty} P(t) = 4\tag{11.124}$$

Alternatively, we can substitute $\theta = 7/16$ in Equation (11.122) to obtain $P = 4$. Figure 11.4 shows P as a function of time when $P(0) = 1$. Note that in this example, since $C = R = 1$, the H_∞ gain K is equal to P .

Figure 11.5 shows the state estimation errors for the time-varying H_∞ filter and the steady-state H_∞ filter. In these simulations, the disturbances w and v were both normally distributed white noise sequences with standard deviations equal to 10. w had a mean of zero, and v had a mean of 10. Both simulations were run with identical disturbance time histories. It can be seen that the performance of the two filters is very similar. There are some differences between the two plots at small values of time before the

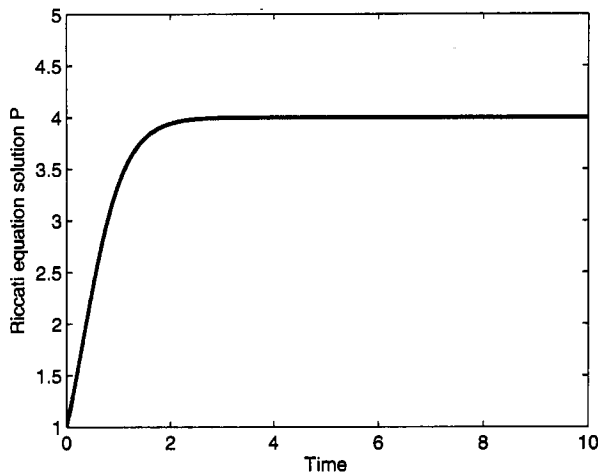


Figure 11.4 Example 11.4 H_∞ Riccati equation solution as a function of time.

time-varying Riccati solution has converged to steady state (note that the time-varying filter performs better during the initial transient). But after the Riccati solution gets close to steady state (after about $t = 1$) the performance of the two filters is nearly identical. This illustrates the possibility of saving a lot of computational effort by using a steady-state filter while giving up only an incremental amount of performance.

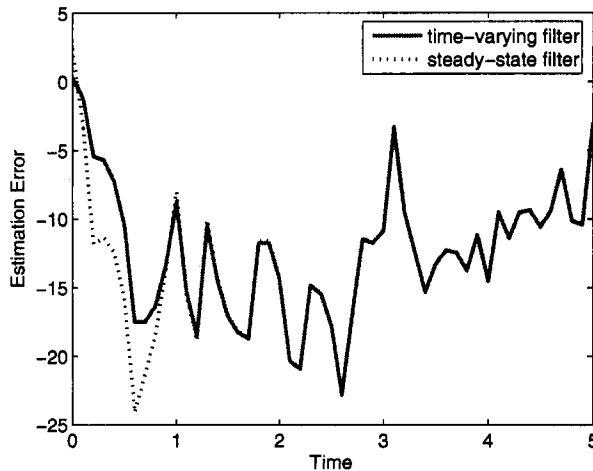


Figure 11.5 Example 11.4 time-varying and steady-state H_∞ filter performance when the measurement noise is zero-mean.

If we use the performance bound $\theta = 0$ in this example then we obtain the Kalman filter. The steady-state Riccati equation solution from Equation (11.120) is $(1 + \sqrt{2})$ when $\theta = 0$, so the steady-state Kalman gain $K \approx 2.4$,

which is less than the steady-state H_∞ gain $K = 4$ that we obtained for $\theta = 7/16$. From Equation (11.118) we see that this will make the Kalman filter less responsive to measurements than the H_∞ filter, but the Kalman filter should provide optimal RMS error performance. Indeed, if we run the time-varying Kalman filter ($\theta = 0$) then the two-norm of the estimation error turns out to be 26.5. If we run the time-varying H_∞ filter ($\theta = 7/16$) then the two-norm of the estimation error increases to 30.0.

However, the Kalman filter assumes that the system model is known exactly, the process and measurement noises are zero-mean and uncorrelated, and the noise statistics are known exactly. If we change the simulation so the measurement noise has a mean of 10 then the H_∞ filter works better than the Kalman filter. Figure 11.6 shows the estimation error of the two filters in this case. The two-norm of the estimation error is 112.8 for the Kalman filter but only 94.2 for the H_∞ filter.

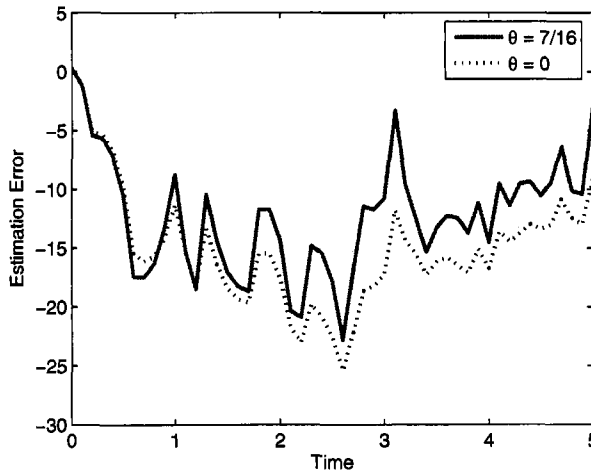


Figure 11.6 Example 11.4 time-varying Kalman and H_∞ filter performance when the measurement noise is not zero-mean.

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As with the discrete-time steady-state filter, if Q , R , and S are all identity matrices, the continuous-time steady-state filter bounds the maximum gain from the noise to the estimation error:

$$\begin{aligned} \|G_{\tilde{z}e}\|_\infty^2 &= \sup_\omega \frac{\|z - \hat{z}\|_2^2}{\|w\|_2^2 + \|v\|_2^2} \\ &\leq \frac{1}{\theta} \end{aligned} \quad (11.125)$$

where ω is the frequency of the noise, and we have defined $\tilde{z} = z - \hat{z}$, $e^T = [w^T \ v^T]^T$, and $G_{\tilde{z}e}$ is the system that has e as its input and $\tilde{z}$ as its output. The continuous-time infinity-norm of the system $G_{\tilde{z}e}$ is defined as follows: